

# Metric-Engineering Propulsion in a Condensate-Based Emergent-Gravity Framework

A Hypothetical Technical Note

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## Abstract

This note develops a purely hypothetical extension of a condensate-based emergent-gravity framework into a vehicle concept that moves by shaping an effective metric rather than by expelling reaction mass. The model is not presented as a claim about real antigravity. Instead, it is formulated as an effective-field-theory thought experiment in which a coherent spin-2 acoustic condensate is driven so as to create a spatially anisotropic effective geometry around a craft. The resulting motion is interpreted as geodesic steering in an engineered metric background. We present a minimal effective action, a control-law interpretation, a spacecraft architecture, and the principal limitations of the construction. The central conclusion is conservative: the framework can describe metric engineering for quasiparticles and possibly for a coupled payload in an analogue medium, but it does not by itself imply reactionless propulsion, gravity shielding, or superluminal travel.

## 1 Introduction

The parent framework treats gravity-like dynamics as an emergent effective theory of a condensed medium. In that language, the metric is not fundamental but is reconstructed from collective degrees of freedom. The present note asks a narrower question: if the effective metric can be shaped, what would a hypothetical “vehicle” look like?

The answer proposed here is deliberately modest. The craft is not a machine that cancels gravity. It is a coherent region of matter in which the local effective geometry is manipulated so that the craft and its attached payload follow a controlled trajectory. In other words, the vehicle is a *metric-engineering system*. Its motion is obtained by tuning the source terms of the effective field equations rather than by pushing on the environment in the usual Newtonian sense.

## 2 Effective model

We use a minimal effective description in which the spin-2 sector is represented by a symmetric perturbation  $h_{ij}^{TT}$  obeying a sourced wave equation of the form

$$\partial_t^2 h_{ij}^{TT} - c_s^2 \nabla^2 h_{ij}^{TT} + m_{\text{eff}}^2 h_{ij}^{TT} = \lambda \Pi_{ij}^{TT} J_{\text{eff}}(x, t), \quad (1)$$

where  $c_s$  is the effective sound speed,  $m_{\text{eff}}$  is the inverse coherence scale of the acoustic sector,  $\Pi_{ij}^{TT}$  denotes the transverse-traceless projection,  $\lambda$  is a coupling constant, and  $J_{\text{eff}}$  is a controllable source density induced by the condensate boundary conditions and internal phase pattern.

The emergent line element is written schematically as

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}^{TT} + \delta h_{\mu\nu}^{\text{shell}}, \quad (2)$$

where  $\delta h_{\mu\nu}^{\text{shell}}$  is a deliberately anisotropic correction generated by the vehicle shell. The shell is the operative part of the device: it creates a spatial gradient in the effective geometry that biases the geodesics of the payload.

**Definition 1** (Metric-engineering regime). *A configuration is said to operate in the metric-engineering regime if the following conditions hold:*

- (i) *the condensate remains phase-coherent over the vehicle volume  $V$ ;*
- (ii) *the backreaction satisfies  $\|h^{TT}\| \ll 1$  so that the EFT is self-consistent;*
- (iii) *the source is varied adiabatically on the timescale  $\omega_{\text{drive}}^{-1}$  with  $\omega_{\text{drive}} \ll c_s/\xi$ ;*
- (iv) *the resulting effective acceleration of the payload is dominated by the gradient of the engineered metric rather than by direct mechanical contact.*

## 3 Vehicle architecture

A concrete hypothetical vehicle has four layers.

### 1. Coherent core

The core is a condensate domain whose role is to maintain a long-lived coherent background state. In the language of the EFT, it provides a stable vacuum in which the effective spin-2 mode can be driven without immediate decoherence. The relevant control parameter is the coherence ratio

$$\chi_{\text{coh}} := \frac{L}{\xi}, \quad (3)$$

where  $L$  is the characteristic size of the coherent region and  $\xi$  is the healing length. The regime of interest is  $\chi_{\text{coh}} \gg 1$ .

### 2. Metric shell

The shell is a thin annulus or layered boundary region where the density, phase, and flow profiles are modulated. Its purpose is to generate an anisotropic source term

$$J_{\text{eff}}(x, t) = J_0(x) + \Delta J(x, t), \quad (4)$$

with  $\Delta J$  chosen so that the effective potential is steeper in the desired direction of travel.

### 3. Navigation controller

The controller is a feedback law that compares the desired trajectory  $X_d(t)$  with the actual centroid position  $X(t)$  of the craft and solves an inverse problem for  $\Delta J$ . A simple variational formulation is

$$\min_{\Delta J} \int_0^T [\|X(t) - X_d(t)\|^2 + \lambda_c \|\Delta J(t)\|^2] dt, \quad (5)$$

where  $\lambda_c$  penalises excessive source strength. This is the mathematically cleanest way to say that the craft “steers” by shaping the metric.

### 4. Isolation layer

The payload region must be isolated from direct turbulent coupling to the shell. In the toy model, the isolation layer is simply a domain where the effective geometry is nearly flat relative to the shell, so the payload experiences a controlled tidal field rather than violent oscillations.

## 4 Propulsion as geodesic steering

If the payload centroid is treated as a test body in the effective geometry, its motion is approximated by the geodesic equation

$$\frac{d^2 X^i}{d\tau^2} + \Gamma_{00}^i (g^{\text{eff}}) c^2 \simeq 0. \quad (6)$$

In the weak-field, slow-motion limit, this becomes

$$\ddot{X}^i \simeq -\partial_i \Phi_{\text{eff}}, \quad (7)$$

with  $\Phi_{\text{eff}}$  the effective potential extracted from  $g_{00}^{\text{eff}} \approx -(1 + 2\Phi_{\text{eff}}/c^2)$ .

Hence the nominal thrust per unit payload mass is not a reaction force in the usual sense, but an effective acceleration

$$a_{\text{app}}^i = -\partial_i \Phi_{\text{eff}}. \quad (8)$$

The engineering problem is therefore an inverse-design problem: choose  $\Delta J$  so that  $\Phi_{\text{eff}}$  has the desired gradient profile along  $X_d(t)$ .

**Proposition 1** (Operational meaning of “antigravity”). *Within this model, “antigravity” does not mean a negative gravitational mass or a cancellation of the Earth’s field. It means that the craft sits in a region where the effective geometry is shaped so that the payload follows a trajectory with reduced apparent weight, altered fall lines, or lateral displacement relative to the ambient background.*

## 5 Scaling, stability, and limits

The metric-engineering regime is only plausible when the field is weak enough for the EFT to remain under control. A useful dimensionless expansion parameter is

$$\epsilon_h := \|h^{TT}\|, \quad \epsilon_h \ll 1. \quad (9)$$

The driving frequency should also remain below the acoustic cutoff,

$$\omega_{\text{drive}} \ll \frac{c_s}{\xi}, \quad (10)$$

so that the condensate responds adiabatically and does not shed energy into uncontrolled high-frequency modes.

A crude estimate of the control power is obtained from the quadratic functional

$$P_{\text{ctrl}} \sim \int d^3x \left[ \dot{\Delta} J^2 + c_s^2 |\nabla \Delta J|^2 + m_{\text{eff}}^2 (\Delta J)^2 \right], \quad (11)$$

which makes explicit that the cost rises quickly with sharper gradients and faster manoeuvres.

Three limits should be stated clearly.

- (a) **No reactionless drive.** The model does not eliminate conservation laws; it reconfigures the way momentum is exchanged between the craft, the condensate, and the effective geometry.
- (b) **No superluminal motion.** All characteristic propagation speeds remain bounded by the relevant effective causal structure.
- (c) **No fundamental antigravity.** The construction is a metric-engineering analogue; it does not by itself produce a repulsive gravitational interaction in the fundamental sense.

## 6 What the craft would look like in practice

If one wanted to sketch the device as a conceptual engineering object, the most economical image is a *toroidal coherent core* surrounded by a *phase-programmed shell*. The shell would continuously rewrite the local effective metric. The vehicle would move by biasing the geodesics around it, much as a fish moves through water not by changing the laws of hydrodynamics, but by shaping the flow field around its body.

The pilot's interface would not be a throttle in the ordinary sense. It would be a control panel for the source term  $\Delta J$  and for the desired metric profile  $\Phi_{\text{eff}}(x, t)$ . Steering would amount to selecting a target geodesic bundle and then solving the inverse problem for the shell configuration that realises it.

## 7 Conclusion

The hypothetical spacecraft described here is best understood as a *metric-engineering platform* built on top of a condensate-based emergent-gravity EFT. Its propulsion law is geodesic steering in an engineered effective geometry, not cancellation of gravity. The model is technically coherent as a thought experiment and is consistent with the conservative status of the parent PDF: it remains an EFT construction, not a completed theory of fundamental antigravity.

The main scientific value of the construction is conceptual. It makes precise what a gravitationally inspired vehicle would mean in an emergent-metric setting, and it exposes the hard boundaries: coherence, stability, backreaction, and the acoustic cutoff. Those are

the real bottlenecks. If any practical analogue were ever built, it would likely be a device for manipulating effective geometry in a condensed medium rather than a machine that defeats gravity itself.

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## **Informal context**

This note is intended as a hypothetical extension of an emergent-gravity manuscript. It is not an experimental claim and does not assert that antigravity is physically achievable with current technology.